Propositions and truth tables

Class 5 of 24: Propositions and truth tables

CSE 2213 Discrete Mathematics (Fall 2026), section AM

Muhammad Anwarul Azim
Lecturer, Dept. of CSE
United International University

True or false?

true or false		no truth value	
Washington, D.C., is the capital of the United States.	**T**	What time is it?	a question
Chattogram is the capital of Bangladesh.	**F**	Read this carefully.	an instruction
1 + 1 = 2	**T**	☐ + 1 = 2	a gap
2 + 2 = 3	**F**	5 + 1 = 2	**F**

A proposition is a sentence that is true or false, but not both.

Letters for sentences

p

There is a quiz today.

q

I come to class.

		p	q
quiz	I come	T	T
quiz	I stay away	T	F
no quiz	I come	F	T
no quiz	I stay away	F	F

Two sentences, four cases. Always in this order.

Quiz on propositions

Quiz 1 minute

For each sentence write: proposition or not. If it is one, give its truth value.

1 $5 + 7 = 10$

a proposition, false

2 $x + 2 = 11$

not a proposition x has no value

3 Answer this question.

not a proposition an instruction

Rosen 1.1, Exercise 1 (d), (e), (f), p. 13.

Not

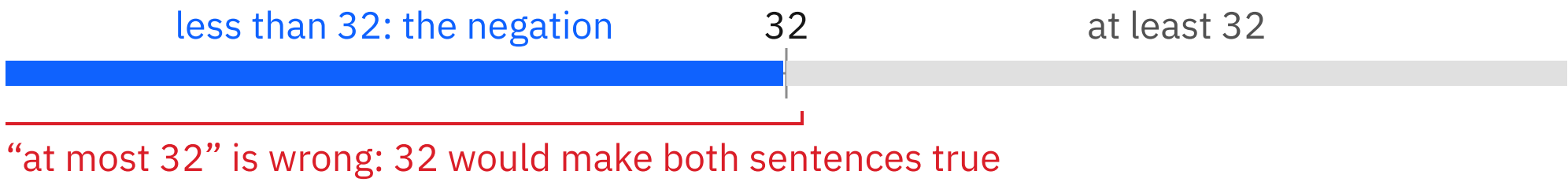
$\neg p$

¬	There is a quiz today.

in English: There is no quiz today.

p	$\neg p$
T	**F**
F	**T**

Rosen, Example 4 (p. 3): “Vandana’s smartphone has **at least 32 GB** of memory.”



And

There is a quiz today **and** I come to class.

There is no quiz today, **but** I come to class.

		p	q	$p \wedge q$	$\neg p \wedge q$
quiz	I come	T	T	**T**	F
quiz	I stay away	T	F	F	F
no quiz	I come	F	T	F	**T**
no quiz	I stay away	F	F	F	F

“But” is “and”.

Or

There is a quiz today **or** I come to class.

I will use all my savings to travel to Europe **or** to buy an electric car.

		p	q	$p \vee q$	$p \oplus q$
quiz	I come	T	T	**T**	F not both
quiz	I stay away	T	F	**T**	**T**
no quiz	I come	F	T	**T**	**T**
no quiz	I stay away	F	F	F	F

“Or” allows both: the inclusive or. “Either … or …, but not both” is the exclusive or.

Quiz on the connectives

Quiz 2 minutes

p: It is below freezing. q: It is snowing. Write 1 and 2 with the connectives; for 3, say which "or" is meant.

1 It is below freezing but not snowing.

$p \wedge \neg q$ "but" is "and"

2 It is either snowing or below freezing (or both).

$p \vee q$ "or both" says inclusive

3 "Coffee or tea comes with dinner."

exclusive you get one of the two

Rosen 1.1, Exercises 13 (b), 13 (d) and 21 (a), pp. 14 to 15.

When is a promise broken?

"If I am elected, then I will lower taxes."

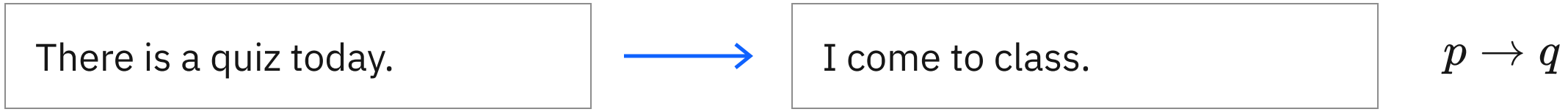
		p	q	$p \to q$	
elected	taxes lowered	T	T	**T**	kept
elected	taxes not lowered	T	F	**F**	**broken**
not elected	taxes lowered	F	T	**T**	nothing to test
not elected	taxes not lowered	F	F	**T**	nothing to test

The conditional $p \to q$ is false in one case only: p true and q false.

Hypothesis and conclusion

If there is a quiz today, **then** I come to class. I come to class **when** there is a quiz today.

hypothesis *conclusion*



		p	q	$p \to q$
quiz	I come	T	T	T
quiz	I stay away	T	F	**F** broken
no quiz	I come	F	T	T
no quiz	I stay away	F	F	T

The “if” part is the hypothesis, wherever English puts it.

A promise without a reason

hypothesis

$1 + 1 = 3$ **F**



conclusion

$2 + 2 = 5$ **F**

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

F? the common answer

T the table's answer: the hypothesis is false, so nothing was promised

A false hypothesis makes the conditional true, whatever the conclusion.

Quiz on the conditional

Quiz 2 minutes

1 and 2: true or false? 3: write it in symbols, with p: You drive over 65 miles per hour, q: You get a speeding ticket.

1 If $1 + 1 = 2$, then $2 + 2 = 5$.

false the one false row: true hypothesis, false conclusion

2 If monkeys can fly, then $1 + 1 = 3$.

true the hypothesis is false

3 You will get a speeding ticket if you drive over 65 miles per hour.

$p \to q$ "q if p": the "if" part is the hypothesis

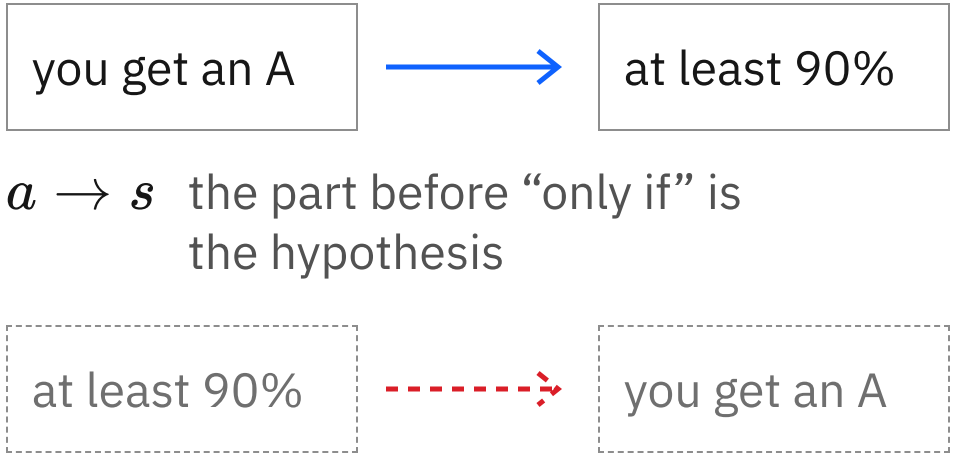
Rosen 1.1, Exercises 19 (a), 19 (d) and 15 (c), pp. 14 to 15.

Only if

"You can receive an A in the course **only if** your score on the final is at least 90%."

a: You receive an A. s: Your score on the final is at least 90%. (Rosen, p. 7)

		a	s	$a \to s$
an A	at least 90%	T	T	**T**
an A	under 90%	T	F	**F** broken
no A	at least 90%	F	T	**T** allowed
no A	under 90%	F	F	**T**

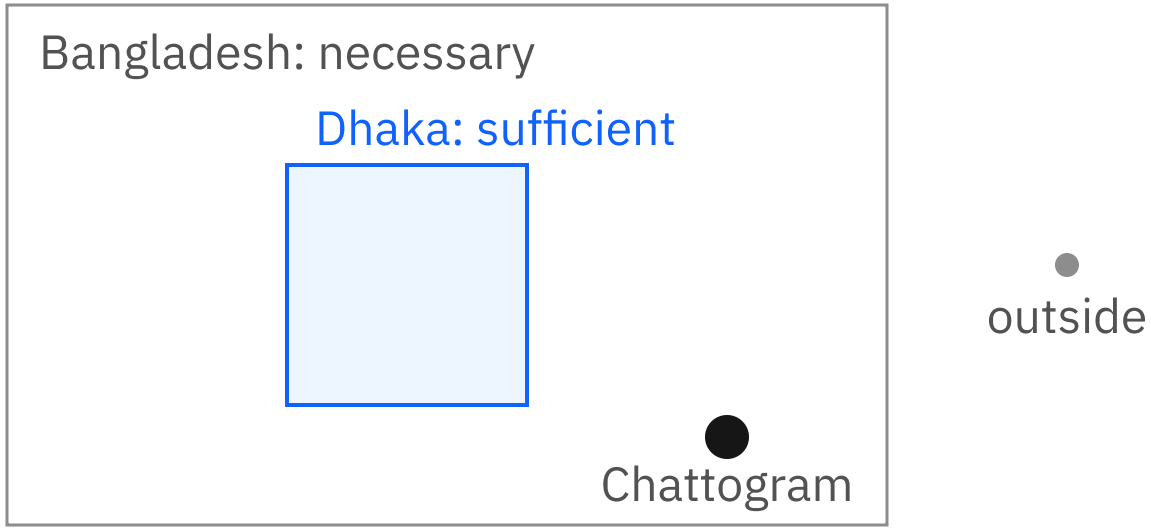


$a \to s$ the part before "only if" is the hypothesis

wrong: 90% is not promised an A (row 3)

"X only if Y" is X $\to$ Y.

Necessary and sufficient



sufficient

I am in Dhaka

→

necessary

I am in Bangladesh

If I am in Dhaka, then I am in Bangladesh.

Being in Dhaka is **sufficient** for being in Bangladesh.

Being in Bangladesh is **necessary** for being in Dhaka.

In Chattogram: in Bangladesh, not in Dhaka. The necessary condition alone is not enough.

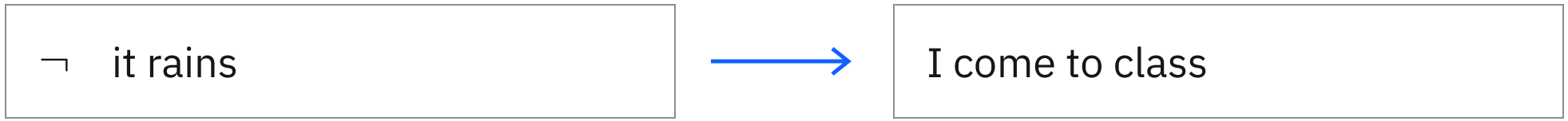
The sufficient condition is the hypothesis. The necessary condition is the conclusion. One arrow.

Unless

"I come to class **unless** it rains."

q: I come to class. r: It rains.

hypothesis | *conclusion*



If it does not rain, then I come to class. $\neg r \to q$

F no rain, and I stay away: **broken**

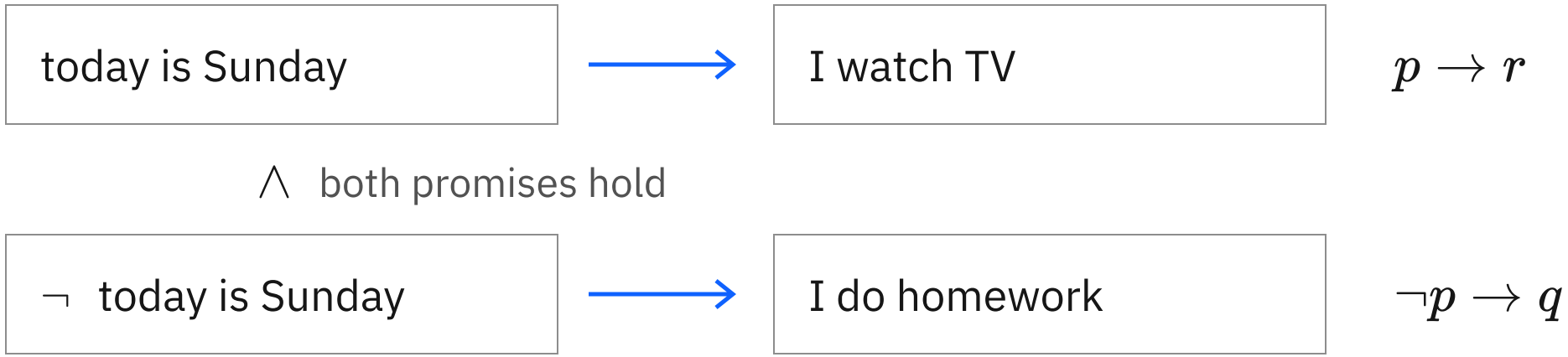
T it rains, and I come anyway: not broken, nothing was promised for a rainy day

"q unless r" is $\neg r \to q$. Unless means "if not".

Otherwise

"If today is Sunday, then I watch TV; **otherwise** I do homework."

p: Today is Sunday. q: I do homework. r: I watch TV. (Summer 2025 mid-term, question 1 (a))



$$(p \to r) \wedge (\neg p \to q)$$

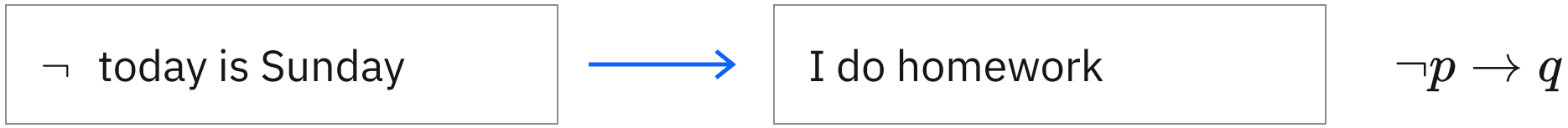
"Otherwise" makes a second promise for the other case.

Except

“I do homework every day **except** Sunday.”

p: Today is Sunday. q: I do homework. (Summer 2025 mid-term, question 1 (a))

Sat	~~Sun~~	Mon	Tue	Wed	Thu	Fri



The answer to give is $\neg p \rightarrow q$: one mark, one formula. The full meaning adds: no homework on Sunday, $p \rightarrow \neg q$.

“Except Sunday” is “if it is not Sunday”.

The idiom sheet

the English idiom	the symbols	shape
if p, q · q if p · q when p · q whenever p · p implies q	$p \to q$	one arrow
p only if q	$p \to q$	one arrow
p is sufficient for q · a sufficient condition for q is p	$p \to q$	one arrow
q is necessary for p · a necessary condition for p is q	$p \to q$	one arrow
q unless r	$\neg r \to q$	one arrow, with a “not”
q every day except Sunday	$\neg s \to q$	one arrow, with a “not”
if p then q, otherwise r	$(p \to q) \wedge (\neg p \to r)$	two arrows
either p or q, but not both	$p \oplus q$	exclusive or
p if and only if q · p is necessary and sufficient for q	$p \leftrightarrow q$	two arrows, one promise each

Quiz on the idioms

Quiz 2 minutes

1 and 2: rewrite as “If ..., then ...”. 3: in symbols, with p: You visit the amusement park, z: You visit the zoo. 4: choose.

1 Jan will go swimming unless the water is too cold.

If the water is not too cold, then Jan will go swimming.

2 Your guarantee is good only if you bought your CD player less than 90 days ago.

If your guarantee is good, then you bought your CD player less than 90 days ago.

3 You can either visit the amusement park or the zoo, but you cannot visit both of the places.

$p \oplus z$ “but not both”

4 “I come to class unless it rains.” Which one: $\neg r \to q$, $r \to \neg q$, $q \leftrightarrow \neg r$?

$\neg r \to q$ the second forbids a rainy-day visit; the third promises twice

Rosen 1.1, Exercises 25 (h) and 25 (g), p. 15; Spring 2025 mid-term, question 1 (b) (i).

Turn the promise around

"The home team wins **whenever** it is raining."

the statement $p \to q$

If it is raining, then the home team wins.

converse: swap $q \to p$

If the home team wins, then it is raining.

inverse: negate both $\neg p \to \neg q$

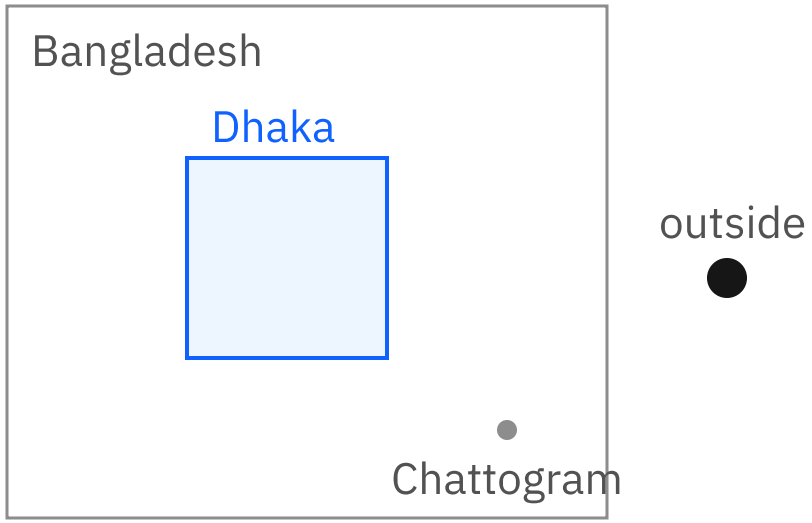
If it is not raining, then the home team does not win.

contrapositive: swap and negate $\neg q \to \neg p$

If the home team does not win, then it is not raining.

Which of the four say the same thing?

The Dhaka test



T the statement
If I am in Dhaka, then I am in Bangladesh.

F converse
If I am in Bangladesh, then I am in Dhaka.

F inverse
If I am not in Dhaka, then I am not in Bangladesh.

T contrapositive
If I am not in Bangladesh, then I am not in Dhaka.

Only the contrapositive always agrees with the statement.

One table for all four

p	q	$p \to q$	$q \to p$	$\neg p \to \neg q$	$\neg q \to \neg p$
T	T	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

- the statement and its contrapositive: the same column in every row
- the converse and the inverse: the same as each other, not as the statement

The same column in every row means **equivalent**.

Quiz on the three forms

Quiz | 2 minutes | Fall 2025 mid-term, question 1 (b), 2 marks

“I come to class when there is going to be a quiz.”

First find the hypothesis and the conclusion; then, as the paper asks, the contrapositive and the inverse.

hypothesis: there is going to be a quiz — the “when” part, at the end

conclusion: I come to class

contrapositive: If I do not come to class, then there is not going to be a quiz.

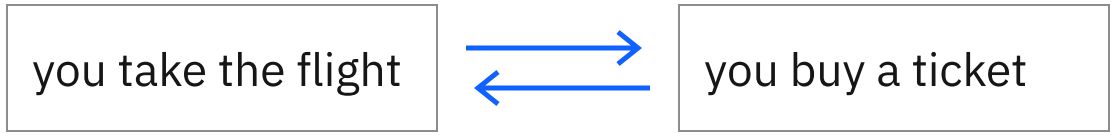
inverse: If there is not going to be a quiz, then I do not come to class.

converse (not asked): If I come to class, then there is going to be a quiz.

If and only if

"You can take the flight **if and only if** you buy a ticket."

		p	q	$p \leftrightarrow q$
flight	ticket	T	T	**T**
flight	no ticket	T	F	**F** a free trip
no flight	ticket	F	T	**F** turned away
no flight	no ticket	F	F	**T**



$p \leftrightarrow q$ is $(p \to q) \wedge (q \to p)$

The biconditional $p \leftrightarrow q$: two arrows, one promise each.

Which connective goes first?



done first … done last

$\neg p \wedge q$ means $(\neg p) \wedge q$

$p \vee q \wedge r$ means $p \vee (q \wedge r)$

$p \vee q \to r$ means $(p \vee q) \to r$

"There is a quiz today and I come to class, or I stay in the hall."

r : I stay in the hall.

$(p \wedge q) \vee r$ **T** $p \wedge (q \vee r)$ **F**

one case tells them apart: no quiz, I stay away, I am in the hall (p F, q F, r T)

When in doubt, write the brackets.

Quiz on symbols

Quiz 1 minute

p: It is below freezing. q: It is snowing.

1 That it is below freezing is necessary and sufficient for it to be snowing.

$p \leftrightarrow q$ necessary and sufficient: both arrows

2 Add the brackets that the order of the connectives gives: $p \vee q \rightarrow r$

$(p \vee q) \rightarrow r$ “or” is done before the arrow

Rosen 1.1, Exercise 13 (g), p. 14, and Table 8, p. 11.

One column at a time

$(p \vee \neg q) \rightarrow (p \wedge q)$

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	**T**
T	F	T	T	F	**F**
F	T	F	F	F	**T**
F	F	T	T	F	**F**

One column per piece, from the inside out. The whole expression last.

Your truth table

Quiz | 4 minutes | Fall 2025 mid-term, question 1 (a), 2 marks

Construct the truth table of $(p \vee \neg q) \rightarrow q$.

p	q	$\neg q$	$p \vee \neg q$	$(p \vee \neg q) \rightarrow q$
T	T	F	T	**T**
T	F	T	T	**F**
F	T	F	F	**T**
F	F	T	T	**F**

The last column equals the q column: the expression is equivalent to q.

What the mid-term asks

Question 1 of the mid-term, 6 to 8 marks, is built from five pieces:

a truth table (two papers of four)

today

converse, inverse, contrapositive (every paper)

today

the idioms: unless, only if, when

today

English sentences into symbols (three papers)

Thursday, online

a chain of named laws (every paper)

Monday

Today in one picture

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \to q$	$p \leftrightarrow q$
T	T	F	T	T	F	**T**	T
T	F	F	F	T	T	**F**	F
F	T	T	F	T	T	**T**	F
F	F	T	F	F	F	**T**	T

the arrow in English

if · when · whenever · only if · sufficient · necessary · unless (not) · except (not)

two arrows: otherwise · exclusive or: but not both

the four forms

$\underline{p \to q}$ $q \to p$ $\neg p \to \neg q$ $\underline{\neg q \to \neg p}$

the statement and its contrapositive agree in every row

Thank you.

Next	Thursday 12 November, online: English into logic
Reading	Rosen 1.1, pages 1 to 13; before Thursday, 1.1 pages 7 to 9 again and 1.2 pages 17 to 18
Practice	Rosen 1.1, exercises 1, 5, 13, 15, 17, 19, 21, 25, 29, 33, 39 (answers at the back)
Moodle	the idiom sheet, tonight



References

Textbook

Rosen, K. H. (2019). *Discrete Mathematics and Its Applications*, 8th ed. McGraw-Hill. Section 1.1, pp. 1 to 13: Examples 1 and 2 (p. 2), 4 (p. 3), 9 (p. 6), the politician's pledge and the English forms of the conditional (p. 7), 12 (p. 9), 13 (p. 10), 14 and Table 7 (p. 11); Tables 1 to 6 (pp. 4 to 10), Table 8 (p. 11). Exercises 1, 13, 15, 19, 21, 25, 29, 33 (pp. 13 to 16), answers pp. S-1 to S-2. Section 1.2, pp. 17 to 18 (translation).

Past papers

CSE 2213 mid-term examinations, Department of CSE, United International University. Spring 2025, question 1 (b) (i); Summer 2025, question 1 (a); Fall 2025, questions 1 (a) and 1 (b).

Examples

The Dhaka sentence: C. A. Gomes, *Propositional Logic, Part 2*, departmental slides, CSE 2213. The quiz and class thread, the Chattogram sentence, the map and the week strip are this course's own.

Symbols used today: $\neg$, $\wedge$, $\vee$, $\oplus$, $\rightarrow$, $\leftrightarrow$, as printed in the book.